

PROBLEM SET 1

ECON301—ECONOMETRICS I

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Remark on notation: The indicator function is denoted $I_A(x)$, recall that this function takes on the value of 1 when $x \in A$ and 0 when $x \notin A$.

1. The following exercise continues the example from Lecture 1. It will also help you remember some important prerequisite concepts from probability and statistics. Let X be a random variable that denotes the annual level of income, in thousands of dollars, from a small town. Let Y denote the annual level of savings, also in thousands of dollars. These random variables follow the following distributions:

$$X \sim \text{Gamma}(2, \lambda)$$

$$Y|X = x \sim \text{Uniform}[0, x]$$

where $\lambda = 0.7$. Recall that the Gamma distribution with parameters (α, β) is given by

$$f(x; \alpha, \beta) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)} I_{[0, \infty)}(x)$$

where $\Gamma(\cdot)$ is the Gamma function. The notation $Y|X = x$ is referring to the conditional random variable with distribution $f(y|x)$ given by the uniform distribution over $[0, x]$, i.e.,

$$f(y|x) = \frac{1}{x} I_{[0, x]}(y)$$

Note, as in the example from the lecture, that by construction we have that $Y \leq X$.

- i. Using the law of total probability, compute the joint distribution of (X, Y)
 - ii. Compute the marginal distribution $f_Y(y)$.
 - iii. Compute the marginal distribution $f_X(x)$.
 - iv. Compute the conditional expectation $E[Y|X = x]$.
 - i. Compute the conditional expectation of Y when $X = 2.5, 5, 10$. Compare with the simulation results from lecture 1.
 - v. Verify the law of iterated expectations for this example by:
 - i. Using the marginal distribution of Y , $f_Y(y)$, compute $E[Y]$.
 - ii. Looking at $E[Y|X]$ as a random variable of X , and using the marginal distribution of X , $f_X(x)$, compute $E[E[Y|X]]$.
 - vi. Compute $\text{Var}(Y|X = x)$.
 - vii. Verify the law of iterated variance for this example by:
 - i. Using the marginal distribution of Y , compute $\text{Var}(Y)$.
 - ii. Using the marginal distribution of X , compute $E[\text{Var}(Y|X)]$ and $\text{Var}(E[Y|X])$.
 - viii. Compute the best linear predictor $E^*(Y|X) = \alpha + \beta X$ by using the formulas in Thm. 2.7.
 - ix. Verify the results from Thm. 2.9 by using items (d) and (h). Conclude that since the conditional expectation is linear, it is also the best linear predictor.
2. Prove Thm 2.4 from Lecture 2. *Hint:* $\text{Cov}(X + Y, X + Y|Z)$.
 3. Let (X, Y) be a r.v. with pdf $f(x, y) = (x + y)1_{[0, 1]}(x)1_{[0, 1]}(y)$, show that

$$E(X^m Y^n) = \frac{(m+1)(n+2) + (m+2)(n+1)}{(m+1)(m+2)(n+1)(n+2)}$$

4. Let $U^* = Y - E^*(Y|X_1, \dots, X_k)$, show that

$$E(U^*|X_1, \dots, X_k) = E(Y|X_1, \dots, X_k) - E^*(Y|X_1, \dots, X_k),$$

and provide an example where the above expectation is different from zero.

5. Let (X, Y) be a random vector with pdf $f(x, y)$. Let $Z = E[Y|X]$. Show that the BLP of Z as a function of X is the same as the BLP of Y as a function of X . Interpret intuitively this result.
6. The daily quantity demanded of unleaded gasoline in a regional market can be represented as $Q = 100 - 10p + E$, where $p \in [0, 8]$ and E is a random variable having a probability density given by $f(e) = 0.025I_{[-20, 20]}(e)$.

Quantity demanded, Q , is measured in thousands of liters, and price, p , is measured in dollars.

- i. What is the probability of the quantity demanded being greater than 70,000 liters if price is equal to \$4? if price is equal to \$3?
 - ii. If the average variable cost of supplying Q amount of unleaded gasoline is given by $C(Q) = \frac{1}{2}Q^{1/2}$, define a random variable that can be used to represent the daily profit above variable cost from the sale of unleaded gasoline.
 - iii. If price is set equal to \$4, what is the probability that there will be a positive profit above variable cost on a given day? What if price is set to \$3? to \$5?
 - iv. If price is set to \$6, what is the probability that quantity demanded will equal 40,000 liters?
7. A newspaper in Nur-Sultan intends to administer two different surveys relating to two different (A and B) anti-tax initiatives. The proportion of surveys mailed that will actually be completed and returned to the newspaper can be represented as the outcome of a bivariate random variable (X, Y) having the density function

$$f(x, y) = \frac{2}{3}(x + 2y)I_{[0,1]}(x)I_{[0,1]}(y).$$

where x is the proportion of surveys relating to the initiative A that are returned, and y refers to the proportion of surveys relating to initiative B that are returned.

- i. Are X and Y independent random variables?
 - ii. What is the conditional distribution of X , given $Y = 0.5$? What is the probability that less than 50 percent of the initiative A surveys are returned, given that 50 percent of the initiative B surveys are returned?
 - iii. Compute the conditional expectation X given $Y = y$, this is a function of y , which is the value that Y can take. Graph this expression. What is the expected proportion of initiative A surveys returned, given that 50 percent of the initiative B surveys are returned?
 - iv. Now compute the best linear predictor of X on Y . Graph this expression and use the best linear predictor to answer the same question as the previous item. Compare your answers for items (c) and (d).
8. Let X, Y and Z be independent random variables with the continuous uniform distribution over $[0, 1]$. Compute the following probability: $\Pr(X < Y < Z)$. *Hint: Come up with a suitable collection of disjoint events whose probabilities add up to 1 and in which one of the events is $\{X < Y < Z\}$.*
 9. Let $Y_1, Y_2, \dots$, be a sequence of independent and identically distributed random variables. Each Y_i represents the amount, in tenge, that a consumer spends at the AX store in Mega Silkway. Let μ_Y and σ_Y^2 denote the common mean and variance of each of the $Y_i, i \geq 1$. Let N be a nonnegative integer-valued random variable that is independent of the sequence $Y_i, i \geq 1$; this random variable represents the number of customers that the AX store receives in a day. Let μ_N and σ_N^2 represent the mean and variance of N . Let T be the total amount spent in a day at the AX store, i.e., $T = \sum_{i=1}^N Y_i$.
 - i. Compute $E[T]$.
 - ii. Compute $\text{Var}(T)$.

Hint: Use the appropriate iterated law for each of the items above.